Magic Cube Power | High-speed file system 3FS

Hardware Design
Software implementation
stress test

High-Flyer, June 13, 2019

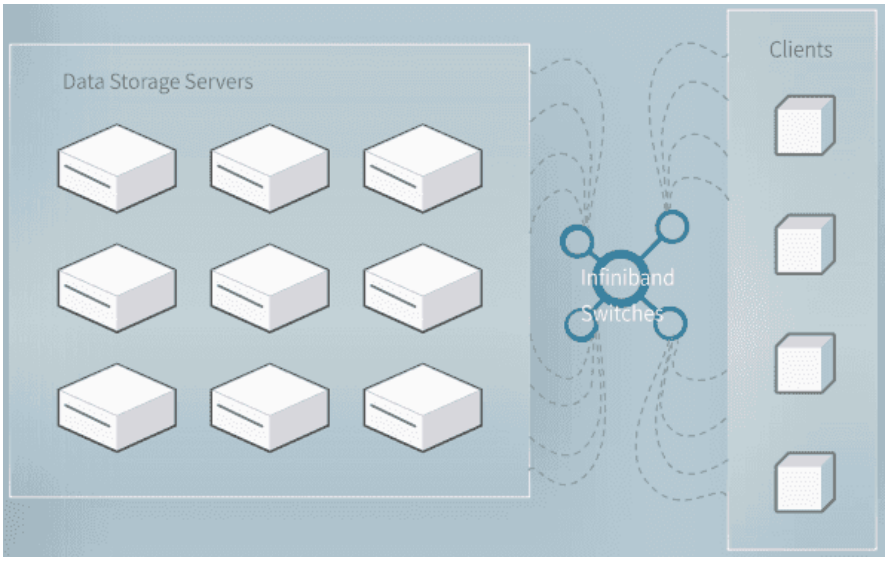
3FS is **a high-speed file system developed by MagicCube AI** . It is an important part of the storage service after the separation of computing and storage in MagicCube "Firefly II". Its full name is Fire-Flyer File System. Because it has three consecutive Fs, it is not easy to pronounce, so it is simply called 3FS.

3FS is a rather unique file system because it's almost exclusively used in AI training scenarios **where models on computation nodes read sample data in batches** , accelerating model training through high-speed computation-storage interaction. This is a large-scale random read task, and the read data won't be used again in a short period. Therefore, we can't use the most important tool, " **read caching ," to optimize file reading; even read-ahead optimization** is useless. Consequently, 3FS's implementation differs significantly from other file systems.

This article will reveal how Magic Square AI designs and implements 3FS, and the resulting acceleration of model training.

Hardware Design

The overall hardware design of the 3FS file system is shown in the following figure:



As can be seen, the 3FS file system comprises two main parts: **data storage services** and **high-speed switches** . The data storage services are separated from the compute nodes and are

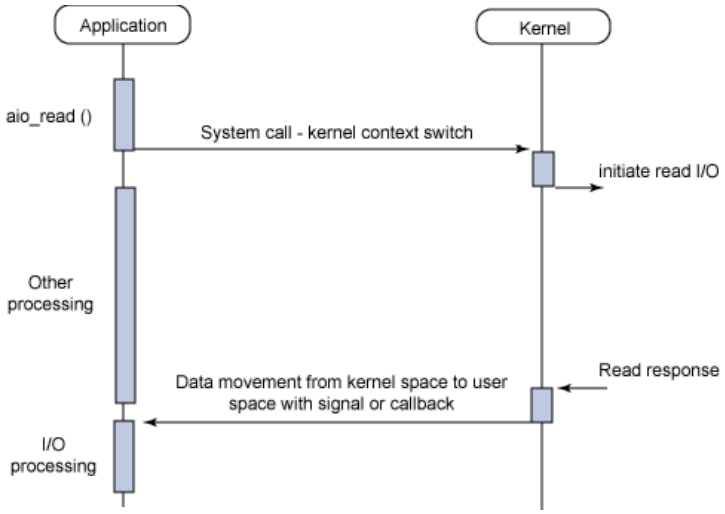
specifically used to store the sample data needed for model training. Each storage service node has 16 15TB SSDs and 2 high-speed network cards, providing strong read performance and high network bandwidth.

Hardware Design
Software implementation
stress test

3FS nodes and compute nodes (Clients) are connected via an 800-port high-speed switch. noting that because one switch connects approximately 600 compute nodes, each compute node can only use one network interface card (NIC). This means that traffic reading sample data from 3FS and other data traffic generated during training (gradient information, data parallelism information, etc.) must share the bandwidth of this single NIC, posing some challenges to the overall read performance of 3FS.

Software implementation

As mentioned earlier, 3FS addresses the specific use case of reading sample data during model training. Unlike common file reading methods, the samples needed for training are random, and often a batch uses unrelated samples. Recognizing this, we decided to design an **asynchronous file reading method** .



Specifically, as shown in the diagram above, 3FS uses Linux-based AIO and io_uring interfaces to read samples. Because file cache is completely useless in 3FS scenarios and instead consumes system memory in a way that is difficult for users to control, affecting the execution of subsequent tasks, we **disabled file cache and only used Direct I/O mode for data reading** . However, it's important to note that in this method, the buffer pointer, offset, and length all need to be aligned. If the user were to perform this alignment manually, it would result in additional memory copying. Therefore, we handled the alignment internally within the file system, optimizing performance and simplifying the process for users.

Using 3FS is very simple for users; they only need to **convert the sample data into FFRecord format** and store it in 3FS. FFRecord is a binary sequence storage format developed by HFAiLab

that is optimized for 3FS performance. It is compatible with PyTorch's Dataset and Dataloader interfaces, making it very convenient to load and initiate training. Project address: https://github.com/HFAiLab/ffrecord

Hardware Design
Software implementation
stress test

When using Magic Square Firefly for model training, you only need to perform feature engi on the raw data and convert it into sample data for input to the model. By loading it through 3FS, you can enjoy the ultimate storage performance experience.

stress test

The Magic Cube Firefly II currently has 64 storage servers forming a 3FS file system. Imagine training a ResNet dataset and reading ImageNet data. The compressed ImageNet file is 148GB, and the decompressed and organized binary training sample data in FFRecord format is over 700GB. Assuming a batch size of 400, fully reading the 40GB memory of a single A100 GPU, using 3FS to read the data, under optimal performance, it only takes about 0.29s to 0.10s per epoch to read the ImageNet data. This greatly reduces the data loading overhead, makes full use of the GPU's computation time, and improves GPU utilization.

```
Epoch: 0, Loss: 5.652562141418457, Acc1: 5.08%, Acc5: 14.80%
cost time per epoch: 247.9163s, read data cost 5.6707s
New Best Acc: 5.08%!
Epoch: 1, Loss: 5.388697147369385, Acc1: 11.29%, Acc5: 27.63%
cost time per epoch: 240.9499s, read data cost 4.4169s
New Best Acc: 11.29%!
Epoch: 2, Loss: 4.034314155578613, Acc1: 21.86%, Acc5: 44.26%
cost time per epoch: 239.5466s, read data cost 4.3452s
New Best Acc: 21.86%!
```

The graph above shows the actual epoch time for distributed ResNet training. As you can see, even under full cluster load, the time spent reading data accounts for only about 1.8% of the total time, demonstrating remarkably good data reading performance.

This article is written by **High-Flyer.**

Return to blog homepage

3FS Optimization 01 | Server-side Optimization →

You may reproduce, excerpt, and quote content from this technical blog without violating the original meaning, but you must comply with the following terms: Attribution—You must credit the original author, but you must not imply in any way that MagicCube endorses you, nor shall it negatively impact MagicCube's rights. Non-commercial Use—You may not use the content of this technical blog for commercial purposes. No Derivatives—If you adapt, transform, or recreate this content, you may not publicly disclose or distribute the modified content; the content is for personal use only.

Contact Magic Square

address 14th Floor, Block A, Huijin International Building, No. 169 Huancheng North Road, Gongshu District, Hangzhou; 45th Floor, East Asia Bank Financial Building, No. 66 Huayuan Shiqiao Road, Pudong New Area, Shanghai

Telephone +86-0571-86656960

Mail info@high-flyer.cn

Zhejiang ICP Registration No. 15029780-1

Zhejiang Public Security Bureau Network Security Permit No. 33010302003447

© All rights reserved by Huanfang Quantitative.